

MATH3303 – Lecture 27

The Correspondence Theorem; Introduction to Euclidean Domains

Review Notes

Reference: Lee, Abstract Algebra, Sections 9.2, 10.2 (cf. Exer. 9.20)

Overview

This lecture establishes a fundamental structural result – the *Correspondence Theorem* for rings – and uses it to compute ideal lattices of quotient rings. It then opens Chapter 10 of the course with the definition of a Euclidean domain.

- **Correspondence Theorem:** a bijection between $\{J \trianglelefteq R : I \subseteq J\}$ and $\{\bar{J} \trianglelefteq R/I\}$ preserving primes and maximals
- **Preimage of an ideal** under a ring homomorphism is an ideal (used in the proof)
- **Applications:** all ideals (prime and maximal) of $\mathbb{Z}_{30}$ and $\mathbb{C}[x]/(x^2 - 1)$
- **Euclidean domains:** definition; the standard examples $\mathbb{Z}$, fields, and $K[x]$

Context. The Correspondence Theorem is the culmination of the ideal/quotient theory from Lectures 22–26. The transition to Chapter 10 then narrows focus to integral domains with extra structure, beginning with the most concrete: Euclidean domains.

1 Correspondence Theorem: Setup

Exercise 1.1: Maximal ideal in $\mathbb{C}[x, y]$ (from lecture)

$(x - \alpha, y - \beta) \trianglelefteq \mathbb{C}[x, y]$ is maximal (continued from Lecture 26).

Theorem 1.1: Correspondence Theorem for rings (from lecture; cf. Lee, Exer. 9.20)

Let R be a commutative ring and I an ideal of R . Then there is a bijection

$$S := \{J \trianglelefteq R : I \subseteq J\} \xrightarrow{\text{bijection}} T := \{\bar{J} \trianglelefteq R/I\}.$$

In addition, this bijection preserves prime ideals and maximal ideals.

Proof. Consider the map

$$\varphi : S \rightarrow T, \quad J \mapsto J/I,$$

and the candidate inverse

$$\psi : T \rightarrow S, \quad \bar{J} \mapsto \pi^{-1}(\bar{J}),$$

where $\pi : R \rightarrow R/I$ is the natural projection and

$$\pi^{-1}(\bar{J}) := \{r \in R : \pi(r) = r + I \in \bar{J}\}.$$

Checks.

- (1) If $J \trianglelefteq R$ with $I \subseteq J$, then $J/I \trianglelefteq R/I$.
- (2) If $\bar{J} \trianglelefteq R/I$, then $\pi^{-1}(\bar{J}) \trianglelefteq R$ and $I \subseteq \pi^{-1}(\bar{J})$. (See Exercise 1.3 below.)
- (3) $\varphi\psi = 1_T$ and $\psi\varphi = 1_S$ (direct verification).

□

Exercise 1.2: Preimage of an ideal is an ideal (from lecture)

Let $\varphi : R \rightarrow S$ be a commutative ring homomorphism, and $J \trianglelefteq S$. Define

$$\varphi^{-1}(J) := \{r \in R : \varphi(r) \in J\}.$$

Then $\varphi^{-1}(J) \trianglelefteq R$.

Proof.

- *Additive subgroup.* For $r, s \in \varphi^{-1}(J)$, $\varphi(r), \varphi(s) \in J$, so $\varphi(r - s) = \varphi(r) - \varphi(s) \in J$. Hence $(\varphi^{-1}(J), +) \leq (R, +)$.
- *Absorption.* For $a \in R$ and $r \in \varphi^{-1}(J)$, $\varphi(ar) = \varphi(a)\varphi(r) \in J$ (since J is an ideal of S). Hence $ar \in \varphi^{-1}(J)$. □

2 Preservation of Prime and Maximal Ideals

Claim. $\varphi : J \mapsto J/I$ preserves prime ideals.

Let $J \trianglelefteq R$ be prime with $I \subseteq J$. By the Third Isomorphism Theorem for rings (Lecture 24),

$$R/J \cong \frac{R/I}{J/I}.$$

Since J is prime, R/J is an integral domain. So $\frac{R/I}{J/I}$ is an integral domain, which is equivalent (Lecture 25, Theorem 2.1) to J/I being prime in R/I .

Similarly, φ preserves maximal ideals (replace “integral domain” by “field” throughout). $\square$

3 Example: All Ideals of $\mathbb{Z}_{30}$

Example 3.1: Ideals, prime ideals, and maximal ideals of $\mathbb{Z}_{30}$ (from lecture)

$\mathbb{Z}_{30} = \mathbb{Z}/30\mathbb{Z}$.

All ideals. Every ideal of $\mathbb{Z}$ is principal, of the form (n) . By the Correspondence Theorem, every ideal of $\mathbb{Z}_{30}$ has the form $(n)/(30)$ with $(30) \subseteq (n)$, i.e. $n \mid 30$. Hence the ideals are

$$(\bar{1}), (\bar{2}), (\bar{3}), (\bar{5}), (\bar{6}), (\bar{10}), (\bar{15}), (\bar{30}) = (\bar{0}).$$

Prime ideals. $(\bar{d})$ is prime $\iff \mathbb{Z}_{30}/(\bar{d})$ is an integral domain. Compute

$$\mathbb{Z}_{30}/(\bar{d}) = \frac{\mathbb{Z}/(30)}{(d)/(30)} \cong \mathbb{Z}/(d) = \mathbb{Z}_d.$$

$\mathbb{Z}_d$ is an integral domain iff d is prime (using the convention $d \geq 2$). Hence the prime ideals are

$$(\bar{2}), (\bar{3}), (\bar{5}).$$

Maximal ideals. $(\bar{d})$ is maximal $\iff \mathbb{Z}_{30}/(\bar{d}) \cong \mathbb{Z}_d$ is a field $\iff d$ is prime. Hence the maximal ideals are

$$(\bar{2}), (\bar{3}), (\bar{5}).$$

4 Example: Maximal Ideals of $\mathbb{C}[x]/(x^2 - 1)$

Example 4.1: Maximal ideals of $\mathbb{C}[x]/(x^2 - 1)$ (from lecture)

By Lecture 26, the maximal ideals of $\mathbb{C}[x]$ are precisely $(x - a)$ for $a \in \mathbb{C}$. By the Correspondence Theorem, the maximal ideals of $\mathbb{C}[x]/(x^2 - 1)$ correspond to maximal ideals of $\mathbb{C}[x]$ containing $(x^2 - 1)$.

We need $(x - a) \supseteq (x^2 - 1) \iff (x - a) \mid (x^2 - 1)$. Since $x^2 - 1 = (x + 1)(x - 1)$, the only linear divisors (up to units) are $x - 1$ and $x + 1$.

Hence there are exactly two maximal ideals in $\mathbb{C}[x]/(x^2 - 1)$:

$$(x - 1)/(x^2 - 1) \quad \text{and} \quad (x + 1)/(x^2 - 1).$$

5 Euclidean Domains: Definition and First Examples

We now begin Chapter 10 of the course: *special types of integral domains*. The first class is the most concrete.

Definition 5.1: Euclidean function and Euclidean domain (from lecture; cf. Lee, Defs. 10.3, 10.4)

Let R be an integral domain. A *Euclidean function* on R is a map

$$\varepsilon : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$$

such that

- (1) for each $a, b \in R \setminus \{0\}$, $\varepsilon(a) \leq \varepsilon(ab)$;
- (2) for each $a, b \in R \setminus \{0\}$, there exist $q, r \in R$ such that

$$a = bq + r,$$

where $r = 0$ or $\varepsilon(r) < \varepsilon(b)$.

The pair (R, ε) is called a *Euclidean domain*.

Example 5.1: The integers (from lecture; cf. Lee, Example 10.5)

$R = \mathbb{Z}$ with $\varepsilon(a) := |a|$ for all $a \in \mathbb{Z} \setminus \{0\}$ is a Euclidean domain.

- $|ab| = |a||b| \geq |a|$ for nonzero a, b .
- The division algorithm (MATH2301) provides q, r with $a = bq + r$, $0 \leq r < |b|$.

Example 5.2: Any field (from lecture; cf. Lee, Example 10.6)

Let K be a field. Define $\varepsilon(x) := 1$ for all $x \in K \setminus \{0\}$.

- For all $x, y \in K \setminus \{0\}$, $\varepsilon(xy) = 1 \geq 1 = \varepsilon(x)$.
- For all $x, y \in K \setminus \{0\}$, take $q = x^{-1}y$ and $r = 0$. Then $y = x(x^{-1}y) + 0$. Both q and r are unique.

Example 5.3: Polynomial rings over a field (from lecture; cf. Lee, Example 10.7)

Let K be a field. Then $K[x]$ with $\varepsilon(f(x)) := \deg(f(x))$ is a Euclidean domain.

- For nonzero $f(x), g(x) \in K[x]$,

$$\deg(f(x)g(x)) = \deg(f(x)) + \deg(g(x)) \geq \deg(f(x)).$$

- The polynomial division algorithm (MATH2301) provides $q(x), r(x)$ with $f(x) = g(x)q(x) + r(x)$, where $r(x) = 0$ or $\deg(r(x)) < \deg(g(x))$.

Remark 5.1: Looking ahead (from lecture)

The next lecture treats the Gaussian integers $\mathbb{Z}[i]$ as a Euclidean domain in detail – the first nontrivial example. After that we develop gcds, the Euclidean algorithm, and associates.

Checklist of Skills

- ☐ State the Correspondence Theorem for rings.
- ☐ Define the maps $\varphi : J \mapsto J/I$ and $\psi : \bar{J} \mapsto \pi^{-1}(\bar{J})$ and verify they are mutual inverses.
- ☐ Prove that the preimage of an ideal under a ring homomorphism is an ideal.
- ☐ Use the Third Isomorphism Theorem to deduce that primality/maximality are preserved by the correspondence.
- ☐ Enumerate all ideals of $\mathbb{Z}_n$ for small n (e.g. $\mathbb{Z}_{30}$) and identify which are prime/maximal.
- ☐ Determine all maximal ideals of $\mathbb{C}[x]/(p(x))$ by factoring $p(x)$ into linear factors.
- ☐ State the definition of a Euclidean function and a Euclidean domain.
- ☐ Verify that $\mathbb{Z}$, any field K , and $K[x]$ are Euclidean domains.